

DECEBAL DOBRICA

Staff Systems Engineer · Rust · Distributed Runtimes & Low-Latency Data Paths · AI Infrastructure
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PROFILE

Systems engineer with 15+ years shipping production software and over 1M lines of production Rust across four platforms. I take end-to-end ownership of hard systems problems: an embedded event store sustaining 469K events/sec with sub-microsecond recall, a multi-venue trading runtime that holds decisions inside venue tick budgets, and the Rust and WebAssembly SDK on the integrity-critical path of an identity system carrying ~5M verification transactions a year. Author of two open-source Rust libraries.

CORE SKILLS

Rust Tokio, Axum, Tonic gRPC, SQLx, Tauri, wasm-bindgen; lock-free concurrency (crossbeam, DashMap, parking_lot); zero-copy Arrow IPC; SIMD JSON; criterion benchmarks gating CI; LTO and codegen-units=1 release engineering; Edition 2024, unsafe_code forbidden at workspace level

Systems Event sourcing and CQRS, write-ahead logs, Parquet columnar storage, append-only history and replay, latency budgeting, failure domains, graceful degradation, memory-bandwidth-aware design, C FFI and NIF interop

AI infrastructure Embedded LLM agents (rig-core), MCP servers and clients, hybrid retrieval with Tantivy, fastembed and instant-distance, GraphRAG, Neo4j, vector plus full-text search

Platform Kubernetes, AWS and GCP, IaC (Pulumi, Terraform, CloudFormation), OpenTelemetry, trunk-based CI/CD; Go, Python, TypeScript

EXPERIENCE

Senior Software Engineer [iProov](#) London · Nov 2025 – Present

- Hold hard latency budgets with no retry path on the encrypted WebSocket channel carrying ~5M biometric verification transactions per year, including protocol-level and wire-format changes between client SDKs and the verification backend.
- Shipped deepfake detection and liveness defences into regulated identity pipelines at bank and government scale.
- Rebuilt the Rust-to-WebAssembly and JavaScript SDK toolchain (Bun, Biome, buf.build Protobuf codegen), cutting CI and build times by over 30%.

Founder & Principal Rust Engineer [Wolven Tech Advisory Ltd](#) London / remote · 2025 – Present

- Authored AllSource, an embedded Rust event store: ~353K LOC, 469K events/sec sustained with sub-microsecond recall, WAL plus Parquet columnar storage, zero-copy Arrow IPC on the hot path, and hybrid search over Tantivy, fastembed and instant-distance.
- Built a seven-day durability stress harness covering crash recovery and concurrent writers, with 32 integration suites and 893 tracing points so failures are replayed instead of reconstructed from logs.
- Shipped Longhand, a Tauri 2 and Rust desktop app with embedded LLM agents over rig-core, an MCP client and 43 MCP tools: ~145K LOC, 409 tests, 10-crate workspace with architecture boundaries enforced in CI.
- Published monorepo-meta on crates.io: a 2.7MB release binary sustaining 3,000+ log lines/sec in ~5MB of memory.

Technical Lead [Ebury](#) London · Apr 2025 – Sep 2025

- Led 10+ engineers through a cutover to an event-sourced architecture for regulated FX payments at ~2M requests/day, cutting API latency 25% and CI build time 30% while holding sub-second cross-region consistency.
- Introduced a “you build it, you run it” model and cut engineer onboarding time 65% under regulatory constraints.

Senior Developer (part-time) [HedgeX Capital](#) Remote · Nov 2024 – Feb 2026

- Built a Rust and WebAssembly trading engine against concurrent exchange WebSocket feeds, improving scalability 20% with hot-path allocation tuned to keep quote-to-decision inside venue tick cadence.
- ~60K LOC Rust: lock-free order-book snapshots, strict decimal arithmetic with no floating point on P&L paths, positions modelled as event-sourced aggregates reconciling after restart and replay.

Engineering Manager / Software Architect [Tellimer](#) London · Aug 2019 – Mar 2023

- Foundational platform hire: delivered 0-to-1 ML serving on AWS (SageMaker, Bedrock), cut infrastructure cost 75% through serverless-first design, and grew the team to 9 engineers.

Earlier: BrainRocket (Kubernetes event-driven greenfield, 2024–2025), Breakout Clips (Technical Lead, 2023–2024), Flyt (Go and Kubernetes, onboarded McDonald's Canada at 10x volume, 2019), Funeral Zone (Chief Innovation Officer 2015–2016 with Neo4j, Software Architect 2017–2019 with OrientDB), full-stack roles in Bucharest from 2010.

EDUCATION

- B.Sc. Information Technology & Mathematics — Military Technical Academy, Bucharest (2007–2011). Unrestricted UK right to work. Certified Fractional CTO (Tech Leaders). Author of the Rust Systems & Agentic AI newsletter.